

3.4.1 Food spoilage and contamination

3.4.1.1 Microorganisms and enzymes

Content	Students must know and understand	Suggested application and food preparation skills
- the growth conditions for microorganisms and enzymes and the control of food spoilage - bacteria, yeasts and moulds are microorganisms - high risk foods - enzymes are biological catalysts usually made from protein.	- growth conditions for microorganisms: role of temperature, moisture, food and time - control of microorganism growth: temperature control, pH, water availability - high risk foods: ready to eat moist foods, usually high in protein that easily support the growth of pathogenic bacteria and do not require any further heat treatment or cooking - control of enzymic action: blanching of vegetables before freezing, use of acids to prevent enzymic browning.	- Bread making (S4 and S10). - Water based methods using the hob – blanching of vegetables to demonstrate the destruction of enzymes in foods (S6). - Oxidation – eg preventing water soluble vitamin loss when preparing and cooking vegetables (S3, S6 and S2).

3.4.1.2 The signs of food spoilage

Content	Students must know and understand	Suggested application and food preparation skills
- enzymic action - mould growth - yeast action.	- enzymic action: ripening of bananas, browning of some fruits - mould growth: eg on bread and cheese. Recognise the signs of mould growth on foods - yeast action on fruits eg grapes, strawberries and tomatoes.	- Preparing fruit and vegetables – mash, shred, scoop, segment, juice and blanch fruits and vegetables to control enzymic browning (S3). - Preparing fruit and vegetables which sustain yeast and mould growth, wash and chill to prevent their growth. Demonstrate the following techniques: de-seed, de-skin (for example, tomatoes).

3.4.1.3 Microorganisms in food production

Content	Students must know and understand	Suggested application and food preparation skills
the use of microorganisms in food production.	• moulds in the production of blue cheese • yeasts to raise bread • bacteria in yoghurt and cheese production.	Make a bread dough, finish and shape a bread dough for use in flat breads, pizza or calzone (S4 and S10).

3.4.1.4 Bacterial contamination

Content	Students must know and understand	Suggested application and food preparation skills
• the different sources of bacterial contamination • the main types of bacteria which cause food poisoning • the main sources and methods of control of different food poisoning bacteria types • the general symptoms of food poisoning.	Contamination from: • other contaminated foods including the following raw foods: meat, poultry, eggs, seafood and vegetables • work surfaces and equipment • the people cooking • pests • waste food and rubbish • campylobacter • e-coli • salmonella • listeria • staphylococcus aureus.	